PATIENT CARE THEORY 2

UNIT 5, PART 2: Maternal Assessment

Marilyn Niffin BSc, ACP
Professor Georgian College
Patient Care Theory
Winter 2020, 2021, 2023

Learning Objectives

- Identify the normal gestation of pregnancy
- Describe the methods used to calculate gestational age
- Describe the methods used to time contractions
- Identify the pertinent information that should be obtained from the pregnant patient
- Explain the significance of performing various assessments of a woman in labour
- Identify factors that determine when childbirth is imminent

ASSESSMENT OF THE PREGNANT PATIENT

- You will see many pregnant patients
- Very few will deliver
- Few will involve complications with mom or baby (~1%)
- However, you must be mentally and physically prepared to deliver a baby

Maternal history

- Maternal age – younger (>20 yrs and <35 years can be associated with a high risk pregnancy but not absolute)
- Due Date/ Gestational Age
  - If unknown, calculate as 9 months plus 1 week from the first day of the last menstrual period OR
  - Add 1 year – 3 months +7 days
  - (e.g. –LMP May 20, 2020 -> May 20, 2021 –> Feb 20 -> Feb 27, 2021
- Are there multiples? – Need to anticipate resources
- Have you been receiving pre-natal care?

37-40 weeks is full term. Under 37 weeks is premature

past pregnancy complications are often reocurring

Maternal history

- is this your first pregnancy?
  - "gravida" refers to the number of pregnancies
  - the current pregnancy is included in the gravida count
  - Nullipara – never delivered before

could have been pregnant without delivery

- do you have any other children?
  - "para" = # of live births (> 20 weeks gestation)
    - (multiple births – twins/triplets etc. are considered one para)
  - muti-paras (or multip – a woman who has delivered 2 or more times) deliver more quickly than "primips"(a woman who has had only one delivery)
  - Can further differentiate
    - T - total # of term deliveries (after 37 weeks)
    - P - total # of preterm deliveries (20-36 weeks)
    - A - total # of abortions/miscarriages (before 20 weeks)
    - L - total # of living children

after 2 c-sections, third should be c-section due to scarring on uterus being at risk for rupture.

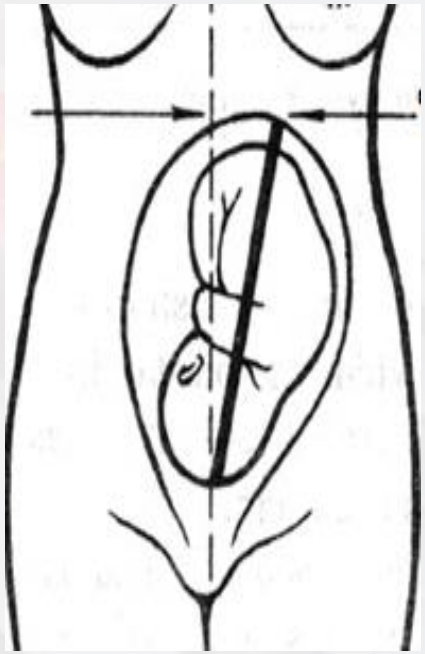
Maternal history

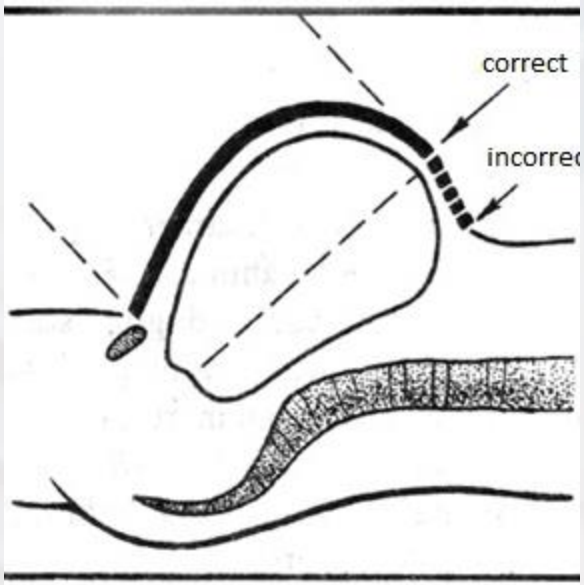
- have you had any complications during this pregnancy? High blood pressure? Diabetes? Swelling? Seizures? Breech? Fetal development?
- Have you ever had a miscarriage of therapeutic abortion?
- What is the current plan for delivery? C-section or natural?
- was your last delivery vaginal or by c-section? Were there any complications with the last delivery (ies)
  - Ask about medical PMHx, Rx and allergies and lifestyle questions - any drugs, smoking, alcohol during pregnancy – need to be sensitive in how these questions are asked
- Is the plan to deliver at hospital (which one) or at home with midwife?

Abdominal Assessment

- You may feel the movement of the fetus – ask mom when she last felt movement
- Estimated gestational age – feel for fundus top of uterus
  - At the symphysis pubis – 12-16wks
  - At the umbilicus ~ 20 weeks gestation (add ~ 1cm for each week of pregnancy)
  - At the costal margin ~ 36+ weeks gestation
- This is important in the case of the unconscious or confused pregnant patient

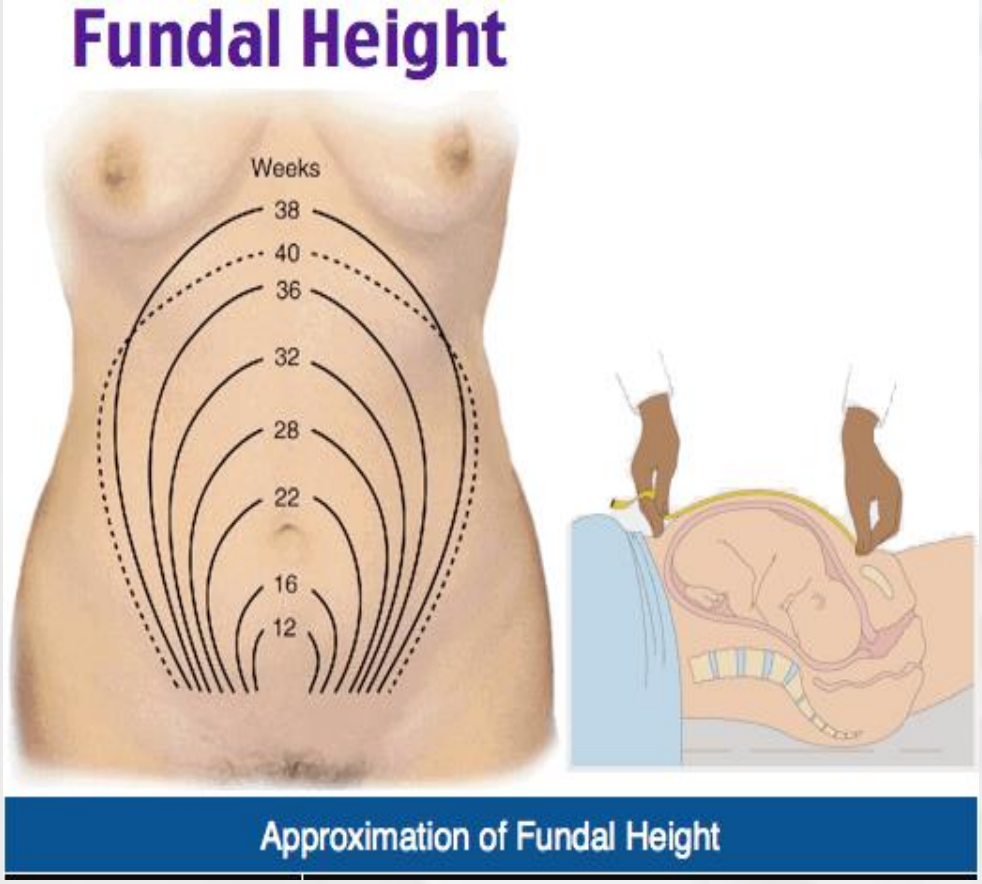
Identifying the Fundus





The Fundus is the top most portion of the uterus

Estimating Gestational Age



Estimated gestational age – feel for fundus

At the symphysis pubis – 12-16wks

At the umbilicus ~ 20 weeks gestation (add ~ 1cm for each week of pregnancy)

At the costal margin ~ 36+ weeks gestation

Incident History

- do you feel the urge to push?
- SROM (spontaneous rupture of membranes)?
  - Large splash vs trickle, colour, odour
- have you had any mucous or bloody discharge?
- Are you having contractions?
- when did the contractions start?
- how far apart are your contractions?
- how long do they last?
- how would you rate the intensity of the contractions 1/10?
- have the contractions been increasing in intensity?

Incident History

- Vaginal bleeding
  - Number of sanitary pads
  - Other vaginal discharges
  - Colour, amount, duration

you can't give analgeseics for imminent birth, only early labour

ASSESSMENT OF THE PREGNANT PATIENT

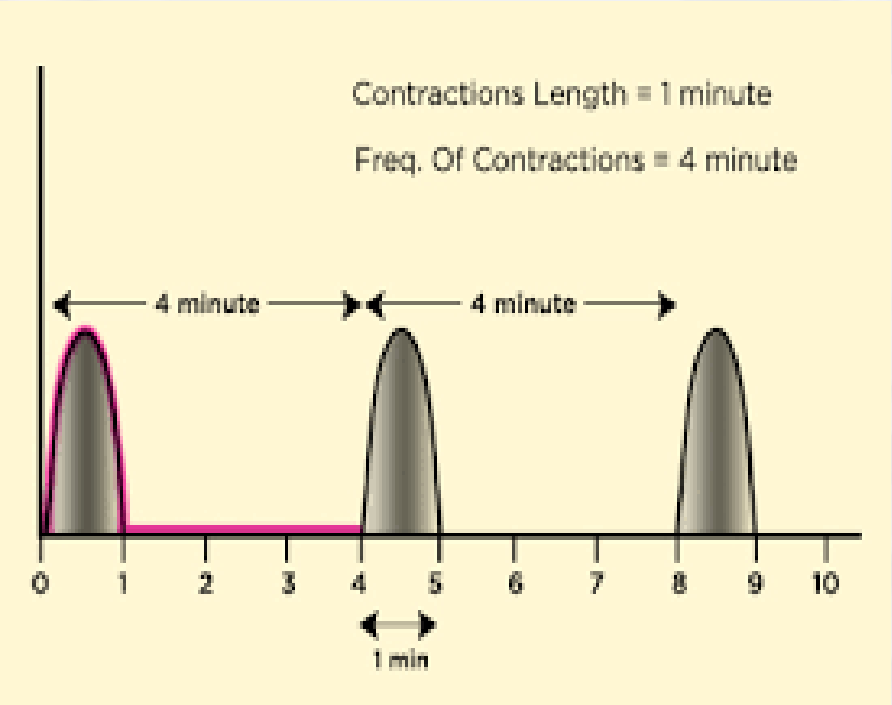
Note

- First pregnancy: you have time **(in most cases)**
- Multiple births = **quicker delivery**

Exam: Mother

- ABCD
- Auscultate the chest
- Vitals (may include BGL if gestational diabetes)
- Focused head to toe
  - e.g. ***Crowning?*** JVD? peripheral edema? Signs of trauma? mucous plug? broken water? Vaginal bleeding?
- +/- auscultate for fetal heart sounds** near impossible
- Keep scene time <u>brief</u> unless birth is imminent

First Stage of Labour



- How are contractions "timed"
- Frequency – length of time from the beginning of one contraction to the beginning of the next
- Length – duration of the contraction
- Strength – how tight is the muscle at the peak of the contraction (fundus)

General Management - transport

move weight off of vena cava

- Position of comfort or left lateral tilt for hypotension
- Always remember that you are caring for two patients, the mother and the fetus
- Monitor vitals, SpO_2 and ECG, IV prn
- Initiate transport ($\leq$ 10 minutes on scene) if birth not imminent
- Monitor frequency and intensity of contractions
- Re-assess for urge to push
- Re-assess for crowning
- Have OBS kit nearby and be ready to open
- If preparing for on scene delivery, prepare all resuscitation equipment as well (for mom and baby)

Management – Stay and Deliver

When do you transport/stay?

- Signs of imminent birth
  - Strong contractions (60-90 secs long)
    - Primips 2-3 min apart; multips 5 min apart
  - Cramping that impedes ambulation
  - Pressure in buttocks, pain or burning in perineum
  - Urge to push, “baby is coming”
  - Bulging membranes/perineum
  - Crowning

Management Stay and Deliver

- prepare to deliver!
- Call for a back up ambulance
- Hope that both patients will be fine; But prepare to resuscitate both

QUESTIONS